

Workshop 1

Julián H. Toro

1. Solve the following inequalities

(a) $|x - 3| < 4$

(b) $\frac{1}{2x - 3} \geq 2$

2. Show that if $|x + 3| < \frac{1}{2}$ then $|4x + 13| < 13$

3. Find the slope of the line through P and Q

(a) $P = (1, 5), \quad Q = (4, 11)$

(b) $P = (-1, 6), \quad Q = (4, -3)$

(c) $P = (-3, 3), \quad Q = (-1, -6)$

4. Show that the lines $2x - y = 4$ and $6x - 2y = 10$ are not parallel and find their point of intersection.

5. Show that the lines $3x - 5y + 19 = 0$ and $10x + 6y - 50 = 0$ are perpendicular and find their point of intersection.

6. Show that the equation represents a circle and find the center and radius.

(a) $x^2 + y^2 - 4x + 10y + 13 = 0$

(b) $2x^2 + 2y^2 - x + y = 1$

7. Under what condition on the coefficients a , b , and c does the equation $x^2 + y^2 + 2ax + 2by + c = 0$ represent a circle? When that condition is satisfied, find the center and radius of the circle.

8. Sketch the region bounded by $y = 3x$ and $y = x^2$

9. Sketch the graph of $x^2 - y^2 - 4x + 3 = 0$